

Counting Objects from Live Video Stream

By

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Counting Objects from Live Video Stream

Major Project Report

Submitted in partial fulfillment of the requirements

For the degree of

Bachelor of Technology in Computer Science & Engineering

By

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Guided By

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CERTIFICATE

This is to confirm that Harshit Lakum's minor project, "Counting Objects from Live Video Stream," was completed under my supervision and assistance and counts toward the partial fulfilment of the requirements for the Nirma University Bachelor of Technology in Computer Science and Engineering degree. The submitted work, in my opinion, has attained the standard necessary for examination.



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Statement of Originality

In accordance with Section 2 of the Indian Contract Act, I certify that the Major Project titled "Counting Objects from Live Video Stream," which I submitted in partial fulfilment of the requirements for a Bachelor of Technology in Computer Science and Engineering from Nirma University, Ahmedabad, contains no material that has been awarded for any degree or diploma in any university or school anywhere in the world, to the best of my knowledge. There has been no attempt to plagiarise this work, and I can vouch for its authenticity. It does not include any previously published or written information other than those citations. I'm aware that if any similarities are uncovered between my work and other published or dissertation work in the future, I'll face harsh disciplinary action.

Signature of Student

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Place: Ahmedabad

Endorsed by

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(Signature of Guide)

ACKNOWLEDGEMENT

Place: Ahmedabad

Date: 04-05-2022,

Dr Vijay Ukani, I want to thank you for your help and assistance in completing my assignment. I appreciate it much.

The chance to complete a project on Counting Items from a Live Video Stream was made possible by our Head of Department, Dr Madhuri Bhavsar, and I'd want to convey my thanks for that. This project would not have been possible without their help and advice.

ABSTRACT

In the age of computers, everything is becoming automated, and one industry is no exception. Image processing and data analytics are automating this industry as well. Early detection is the most effective method for reducing traffic congestion caused by automobiles. Images and videos of automobiles are pre-processed. The Haar-Cascade enhances the image's contrast. Then, using the random walk segmentation approach, the data is broken down into manageable parts. For example, in segmentation the image's ROI is divided, and then its borders are corrected. In a third section, the never-ending pixel shift is divided. In the grouping, the detrimental and non-destructive effects are compared to the model that was already developed. Each of the methods outlined above is able to handle and analysis images and data in a novel way. When it comes to accuracy, cutting-edge XGboost computation will aid in the future since it uses less information but produces higher precision.

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Chapter 1. INTRODUCTION

This project is about Counting Objects from Live video, for that we take one of the Use case of this project and we find an application about Counting Cars with given Video. With use of python programming module OpenCV and machine learning. We use Haar-Cascade for training. In order to track and count we need Object-ID, its Previous Centroids and Whether or not the object has already been counted. The time factor is essential for growth of the psyche. The picture dealing with these can distinguish these threatening developments in the early times of the process. The picture getting ready technique contains four fundamental stages, pre-dealing with, division, including extraction and gathering. While still accurately representing the primary educational assortment, feature extraction is used to minimize dimensionality in a hidden course of action to dynamically plausible social events (Features). There are many ways to approach this, such as making a comparison between feet and meters, or reducing the number of pixels in a picture (moreover named a part vector). Concluding on a subset of the hidden features is known as an "incorporate-decision" choice. We presume that this Project will operate in this manner because we have faith that the selected Features will include the important information from the data, such that the ideal task may be completed using this condensed representation rather than the whole starting data.

Chapter 2. LITERATURE SURVEY

In this project "counting objects from live video," we have to idealise some of the parts, starting with its motive, which we utilize with some of the methods where we use Python-OpenCV and also use Haar-Cascade. Here we use one of the use cases in which we count cars from a given video. For that, we have to go from some of the methods. In priority, we use image tracking using its object id and then we find their objects and then, after completing the detection phase, we count those cars. So here we are using this systematic for counting cars and also this is one of the use cases for this project.

2.1 Image Tracking Using It's Object ID

For Object Tracking, we need to assign every object to their unique ID and then track them by using their unique ID. For example, if there are a bunch of cars, how to assign each car a unique ID? So, for that, we have to give each car a unique ID and then after giving, we can identify and also use it for tracking. It is very necessary to assign a unique ID for image tracking. We can also say that giving a unique ID is just like giving unique identification and, using this scenario, we can track an image because it has unique identification. Many experts have been inspired by the availability of enormous mixed media data sets and the development of data highways to devise effective recovery procedures based on their substance. Computerized picture databases, however, have opened up a technique for satiating based searching and recovery. Recovering the material that is dependent on image highlights like tone, surface, and form has required a great deal of investigation. As part of this paper's effort, we propose to eliminate the waste and significant-level parts from an image using an important input to create a productive picture recovery framework.

2.2 Finding Object's Centroids

The main reason for finding objects centroid is to track them by their previous centroids to their next centroids. Its main purpose is to track that image and also count objects. It is very needed because in video, which is a collection of images, if we want to count objects we have to track them and after finding their centroids, we can say that at the start they have certain centroids and after that they consist of different centroids which give us that object is running and from that we can count that object. So basically, finding centroids is a must needed for counting objects in any

video. If an object is a triangle, it has three vertices, and we get the centroid of the triangle by taking all three x and y, summation of xs and ys, and dividing them by three, but not in every case we get an object whose shape is uncertain, so we take a minimum of 5 vertices which are at the edge of that object's shape and the number of vertices we have. We take that number and we do summation of those vertices xs and ys and divide them with the number of vertices we have. And that's how we found centroids for every object. The main strategy was imagined when it was seen that little growths have a recognizable mark in shape space, where "arch" is the nearby curve of the picture information when seen as a help map.

2.3 After Detection, Counting Objects

The main thing after detection is counting objects. Detecting image or video we identify and locate one or more effective targets from still image or video data. Tracking and counting are two separate processes; tracking is more efficient than simply spotting an item in each frame, while counting is more efficient than simply spotting it in each frame. If there are multiple things in the scene, tracking must be able to identify them across frames. There are times when object detection fails, however this is taken into consideration while tracking. There are a small number of factors that must be taken into consideration while optimizing a movie for counting purposes. We must first turn the image/video into grayscale before reducing it to pure black and white and then masking the image/video and then creating a video sequence. After all this, we can count the number of objects in a movie or picture by reading its frames. On a check arranges framework, the invariant components are used to differentiate between real knobs and a few types of veins that are successfully providing bogus positive findings. OpenCV is used for object identification and tracking, and Haar-Cascade is used for training the dataset. For detection and counting, we create bounding boxes of objects and calculate their respective centroids, and then Euclidean distance between any new centroids and existing ones. After that, when centroids intersect the line from the top of the frame, the in counter is increased, and vice versa when it down the frame out counter is incremented, and after that all things are done .A line we draw in between for counter and counting objects form that video/image.

Chapter 3. SYSTEM ANALYSIS

For efficient object detection and counting, we need an environment in which we use Python-OpenCV and also we can use machine learning methods easily. For that, we need a good system because for detecting and counting, we also use cascade method. For all these things, we need some basic requirements for running programs and working smoothly. For that, we need some basic requirements and those are given below. For hardware, the minimum requirement is given in **Table 3.1** and for software, the minimum requirement is given in **Table 3.2**.

3.1 Proposed System

An efficient object recognition solution based on python OpenCV machine learning and a Cascade function learned from both positive and negative photos, it is an.xml file with several features sets and each xml corresponds to one unique use case. This model is basically count cars from given video and detect them with use of this we can manage traffic on roads.

Advantages:

1. When compared to CNN, it has the ability to simultaneously identify and classify.
2. It do not require changes in their original structure to increase the quality of recognition.
3. It is simpler to build than a CNN because of the parameter variation that allows the number of stages and positive and negative input to be adjusted.

3.2 System Requirements

Here, for system requirements, hardware requirements for this project are given below in **Table 3.1**. The software requirements for this project are given below in **Table 3.2**. They are giving an indication of the requirements of this project.

HARDWARE TOOLS	MINIMUM REQUIREMENTS
Processor	i5 or above
Hardware	10GB
RAM	8GB
Monitor	15.6" Coloured
Mouse	Optical
Keyboard	122 keys

Table 3.1 - Hardware requirements

SOFTWARE TOOLS		MINIMUM REQUIREMENTS
Platform		Windows, Linux or MacOS
Operating System		Windows, Linux or MacOS
Technology		Machine Learning-Python
Scripting Language		Python
IDE		Pycharm IDE

Table 3.2 - Software requirements

Chapter 4. SYSTEM DESIGN

The prime focus of this project is to detect and count objects from video. For that, we use steps to achieve those expectations. Here, we use a Data flow diagram and some UML diagrams for preplanning and executing those things. We are using Python-OpenCV and machine learning modules. We use Cascade for training datasets. After training, it will give us an xml file which we use in further as pertained model. We use an algorithm for detecting cars. After that, for counting, we will use the Separate algorithm, which helps to count cars using the concept of line-separator.

4.1 Data Flow Diagram

For understanding and also planning any software, we use data flow diagrams. As a tool for preplanning and stating how the system will work, as well as an image representation for any working model, **Figure 4.1** shows a data flow diagram for understanding project work. As input, we will give video files to the optimizer and parsers, who will process the video in a justified manner for machine understanding. After that, processed video will go to the object detection module, and after detecting objects, it will go to output, where it will load, transform, and extract information. At the end, it will give the result (final output).

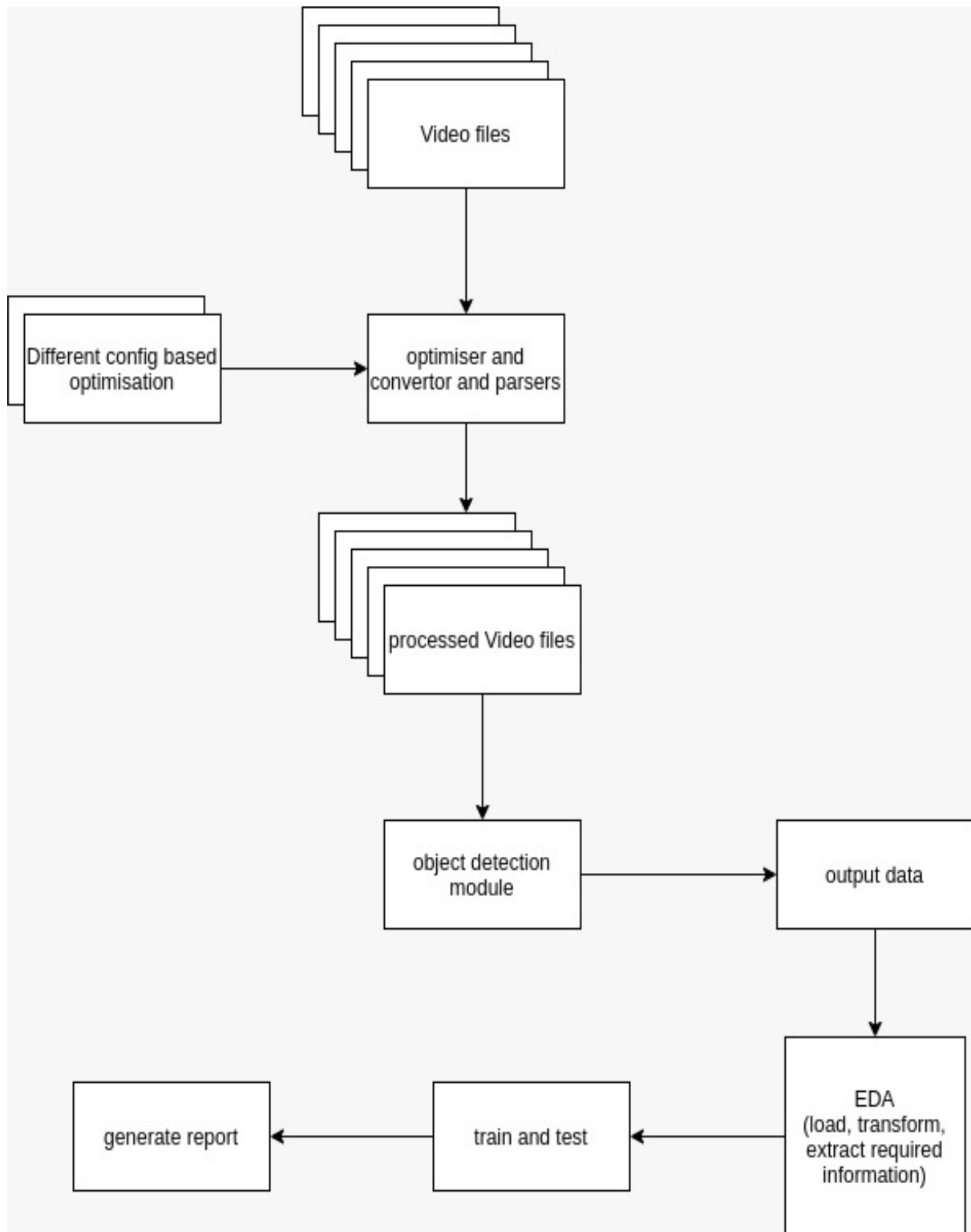


Figure 4.1 - Data Flow Diagram

4.2 UML DIAGRAMS

UML diagrams help us to understand systems with provided standards. There are 14 different types of them. And each has their own specifications. In the categories they divide into, there are two types: one is the structure diagram, and the other is the behavioural diagram. Here we used the activity diagram and block diagram. In **Figure 4.2** there is an activity diagram and in **Figure 4.3** there is a block diagram. An activity diagram is a way to represent in a graphical way what it is easy to guide and understand. On the other hand, we have a block diagram, which is a concept model. It processes gradually.

Activity Diagram:

Here in **Figure 4.2**, there is an Activity Diagram for a given system. The Activity Diagram is one of the UML diagrams. It indicates the system. At first, it selects the target area, after that it will initiate from given image or video data. After that, it will transfer it and then it will do back ground extraction as grey scale. After doing all these, it detects a vehicle from a given input, and from this we will get an output, and by keeping this output, we can use this data in the future for traffic management.

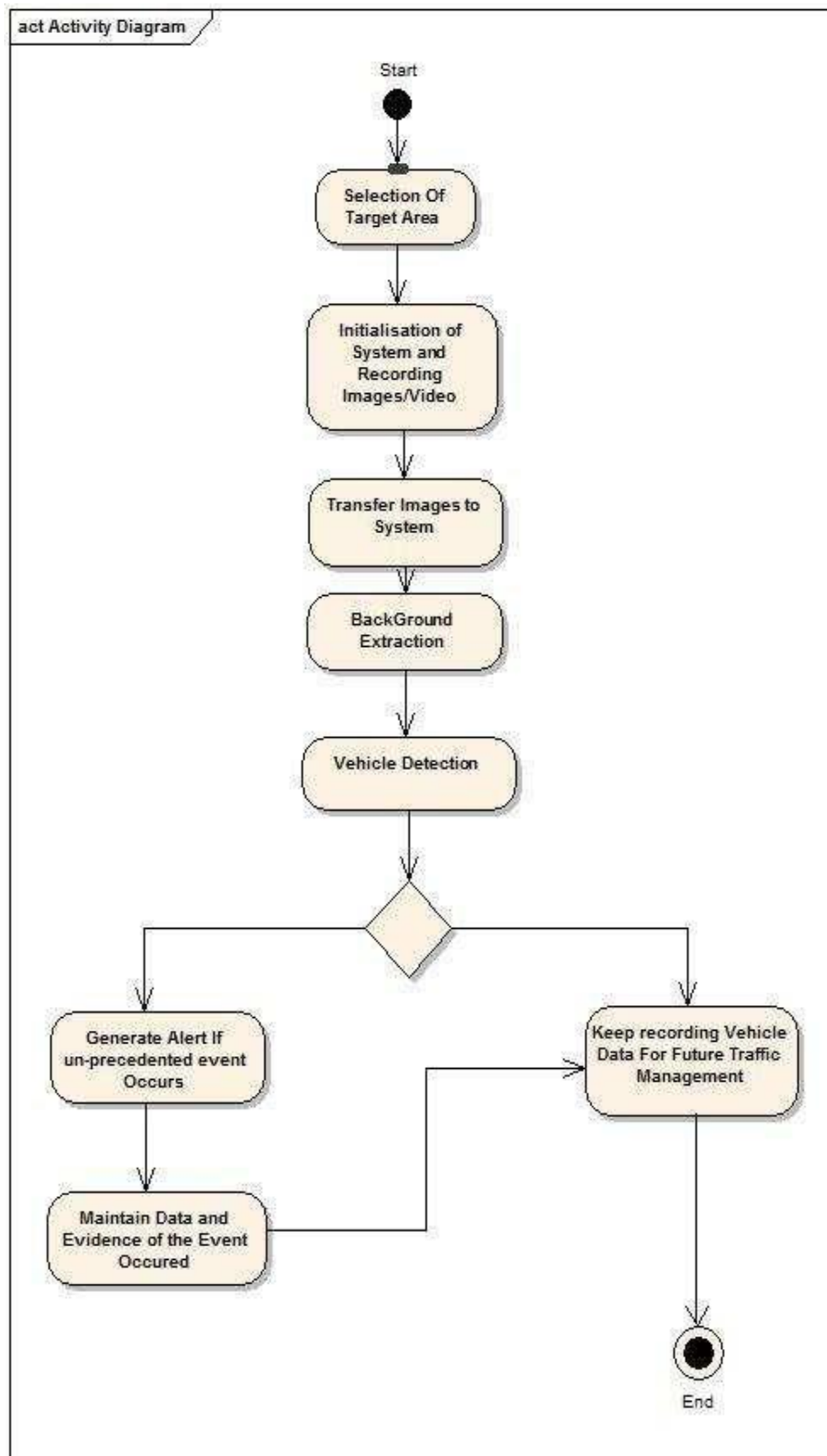


Figure 4.1 - Activity Diagram

Block Diagram:

Here in **Figure 4.3**, there is a Block Diagram for a given system. The Activity Diagram is one of the UML diagrams. As for this system scenario, the block diagram shows the system first receiving video input, then extracting video frames into grey frames, then background subtraction module handling frame handling and detecting objects, and finally classifying and counting objects (cars).

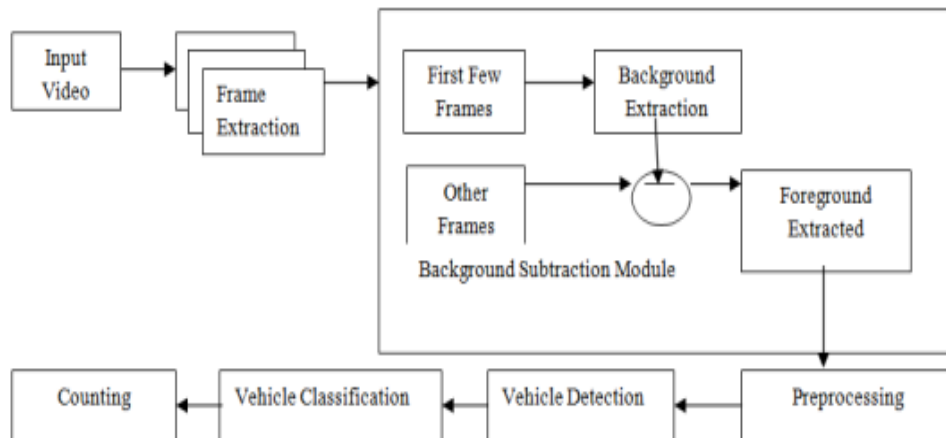


Figure 4.2 - Block Diagram

4.3 IMPLEMENTATION

MODULES:

Upload dataset.

Read & Split, dataset to train & test.

Take sample video for testing

Execute program.

Chapter 5. SOFTWARE ENVIRONMENT

For this project, we need an easy-to-use and accessible software environment. Because for any detection or which includes image or video, we have to need an easy-to-apply software environment, and that's why using Python-OpenCV is a great choice because of its benefits and easy accessibility. We use machine learning and the cascading method for training and testing purposes. It is really needed to make an easily accessible software environment because it helps to achieve sustainability and accuracy.

5.1 Python

We use Python and OpenCV in this program because it is easy to code and it has the advantage of focusing on design. It is also helpful for machine learning. If there is a work-related video or image in which we use machine learning, so python is the easiest to use. If we want to program with OpenCV, so Python-OpenCV is the best to use and also easy to access. It is also portable and great for IOT devices as well, so using Python in this model is easy to understand. It has fast prototyping, vast libraries for machine learning, and is commonly used for machine learning.

5.2 Machine Learning

It is a development of AI and a superset of deep learning that makes use of structured data. Various types of algorithms may be used to model functions and predict future actions based on information. The Haar-Cascade is one of the parts of machine learning for fast output and accuracy. Haar-Cascade is very useful and it only works for image detection. It is easy to use and easy to understand, as well as fast processing, so machine learning for this project was chosen because of its advantages.

Chapter 6. RESULT ANALYSIS

Here, for data analysis for this project, there are some basic uses of cascade methods from where we use positive and negative images for training datasets. After that, we get an xml file. That xml file works as pre-trained. After that, for testing, we take a video sample and at the end we get the detection of objects and the counting of them.

Dataset (positive and negative images): Here, as given below in **Figure 6.1**, there are datasets of positive and negative images. Positive images contain datasets of all cars in various postures, while negative images contain datasets of all other things that do not include cars, such as trees, lamps, dogs, cats, and so on. Dataset is used for making xml files that work as pre-trained datasets. In **Figure 6.2**, which is given below, is the testing video sample.

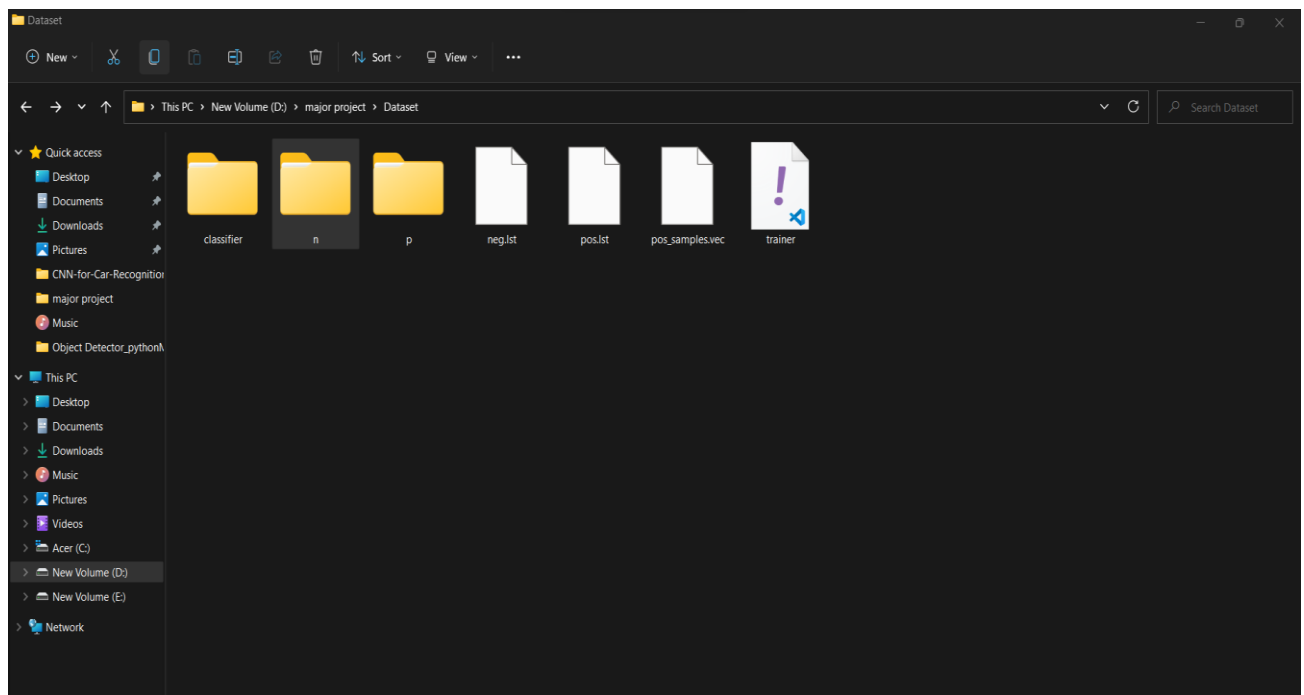


Figure 6.1 - Dataset

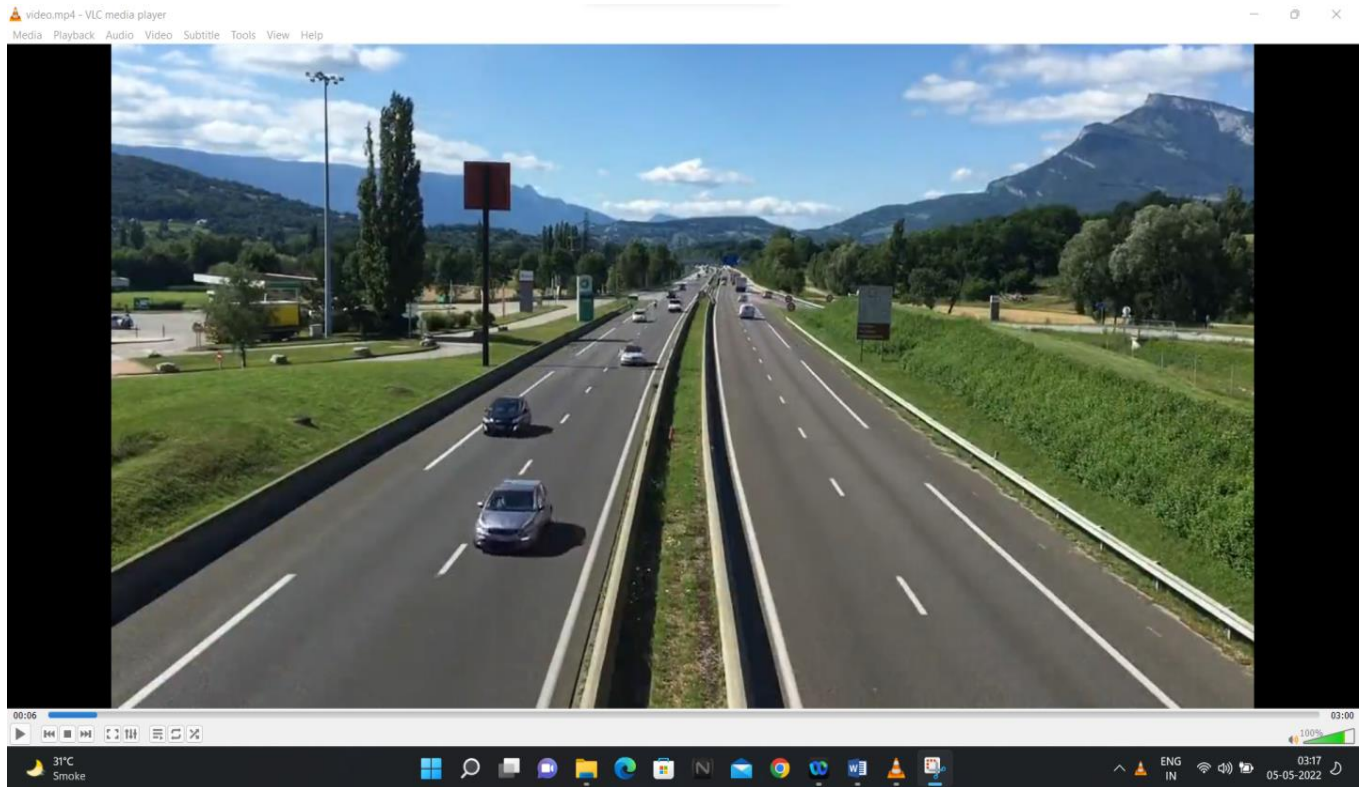


Figure 6.2 - Testing video

Extraction of video in different Frames: Here, as given below in **Figure 6.3**, there is an extraction of testing video in the threshold frame. In **Figure 6.4**, there is an extraction in the Delta Frame. And in **Figure 6.5**, there is an extraction in the grey frame. Extraction of testing video is needed for scaling objects, and using these extractions, we can make detection accurate. And using this, we can get a high level of accuracy for counting objects.

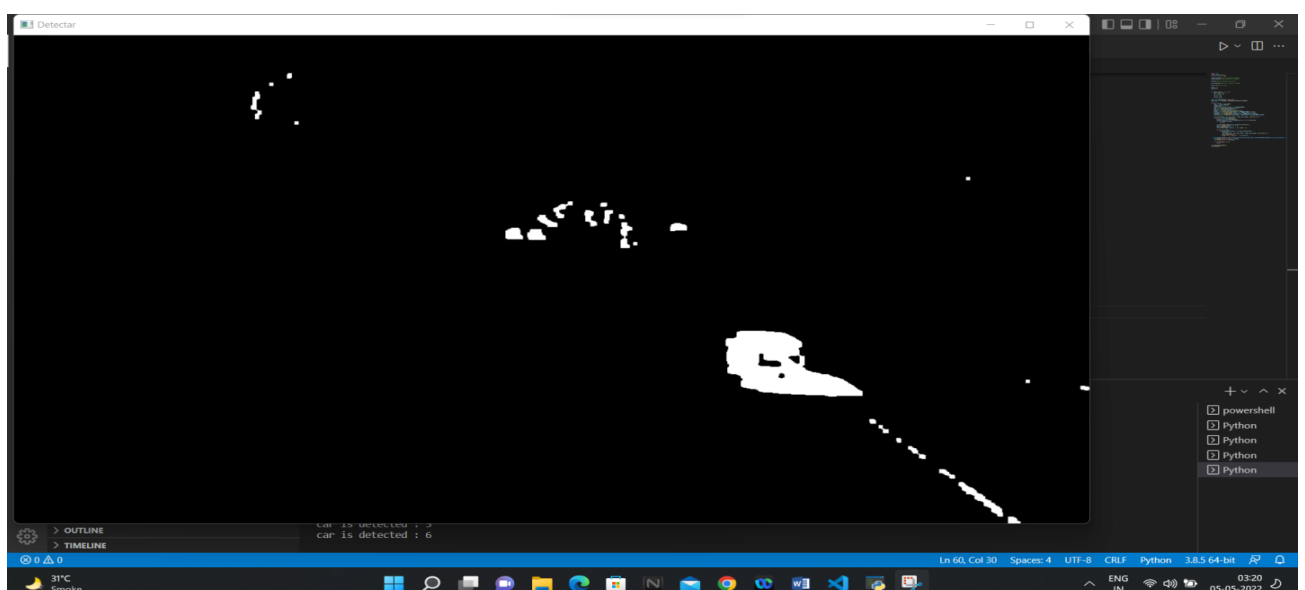


Figure 3.3 - Threshold frame

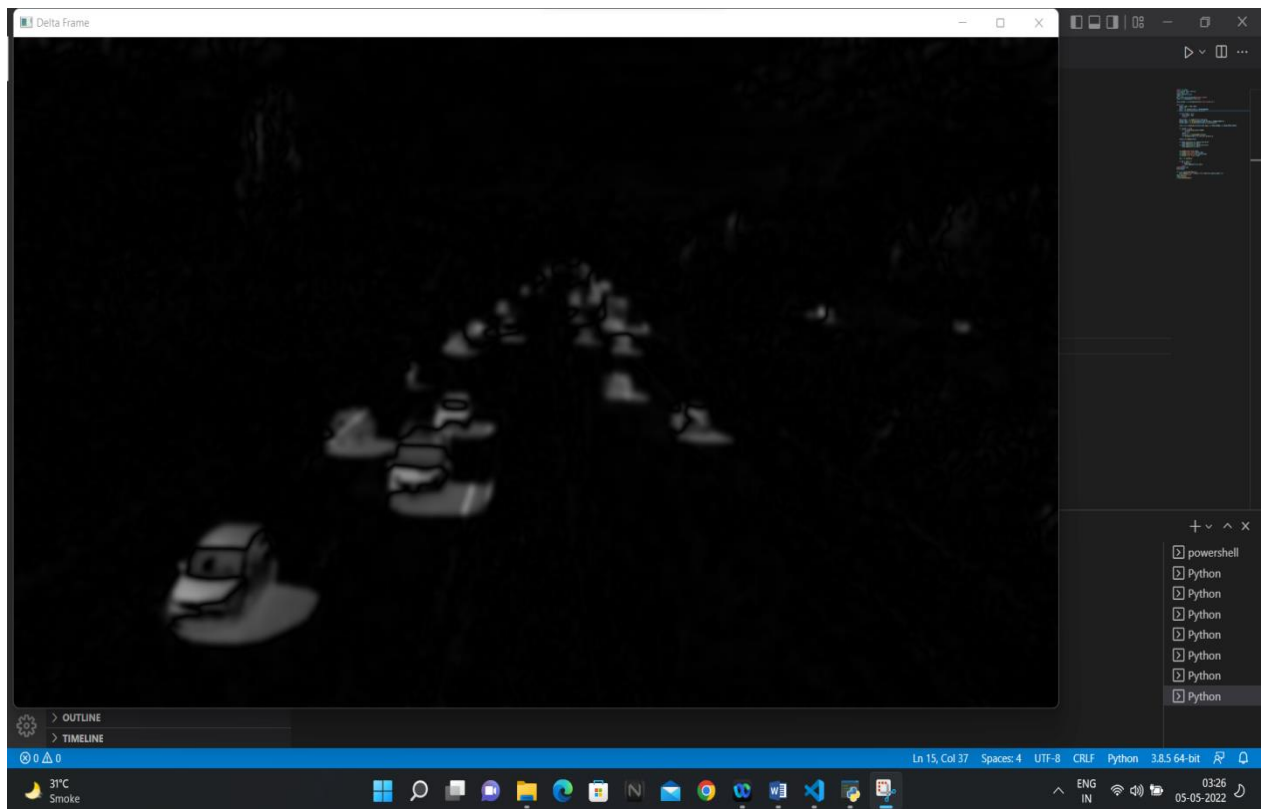


Figure 6.4 - Delta frame

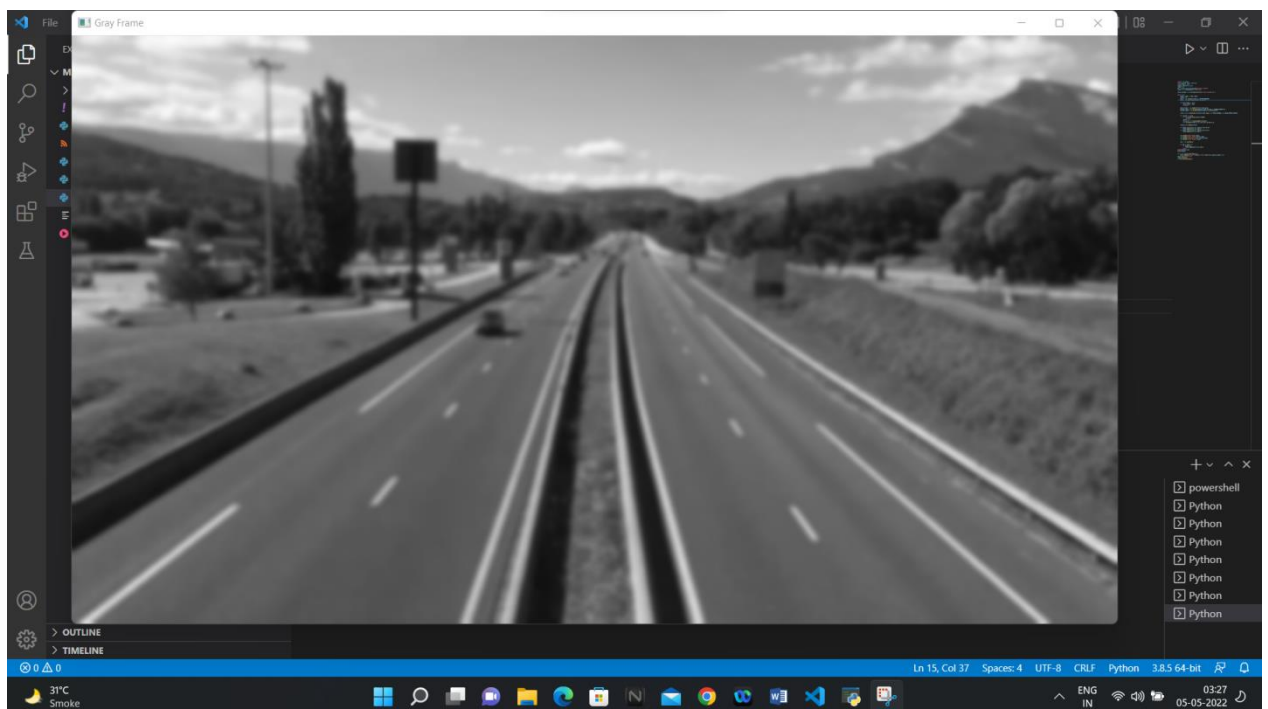


Figure 6.5 - Gray frame

Detecting and Counting Cars, Output: Here, as given below in **Figure 6.6** and **Figure 6.7**, there is an output of video where the purpose of detecting and counting objects has been fulfilled. So, from the start, make a pre-trained dataset using the Cascade method. After that, giving input video for testing, the system extracts input video into different frames. It extracts into Threshold Frame, Delta Frame, and Gray Frame. After extracting, using a separator algorithm, we were able to count that object when it passed through the red line. It would get counted, which is clearly shown in **Figure 6.6** and **Figure 6.7**. And at the end, in **Figure 6.8**, there is a partial part of the code for understating.

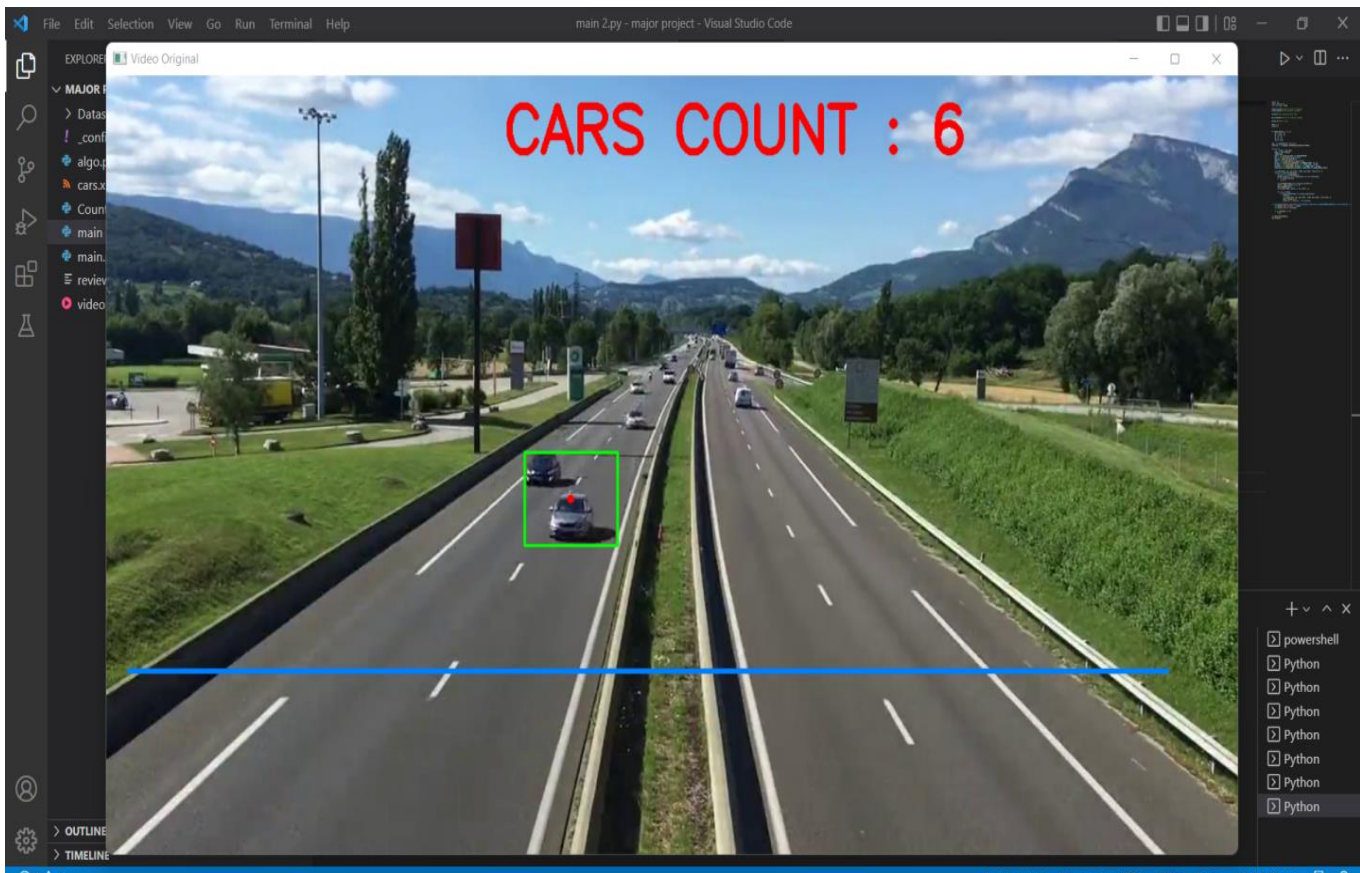


Figure 6.6 - Output

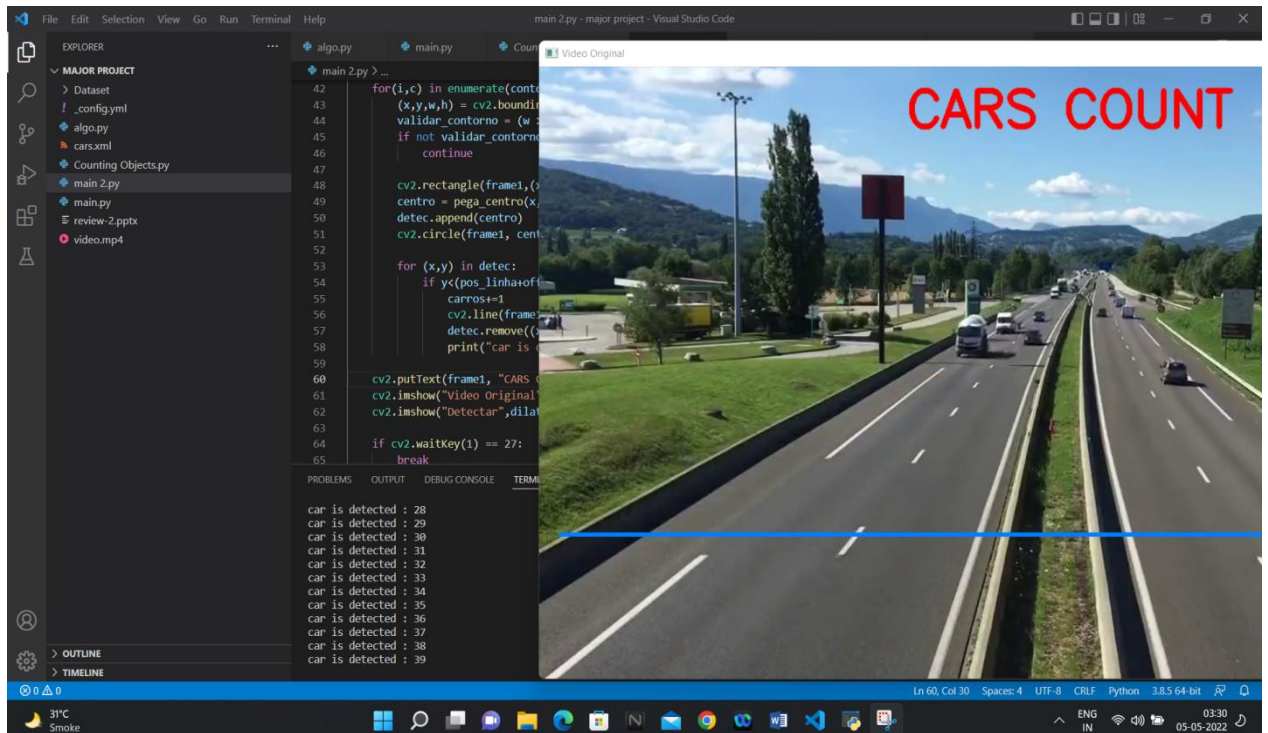


Figure 6.7 – Output, Counting Cars

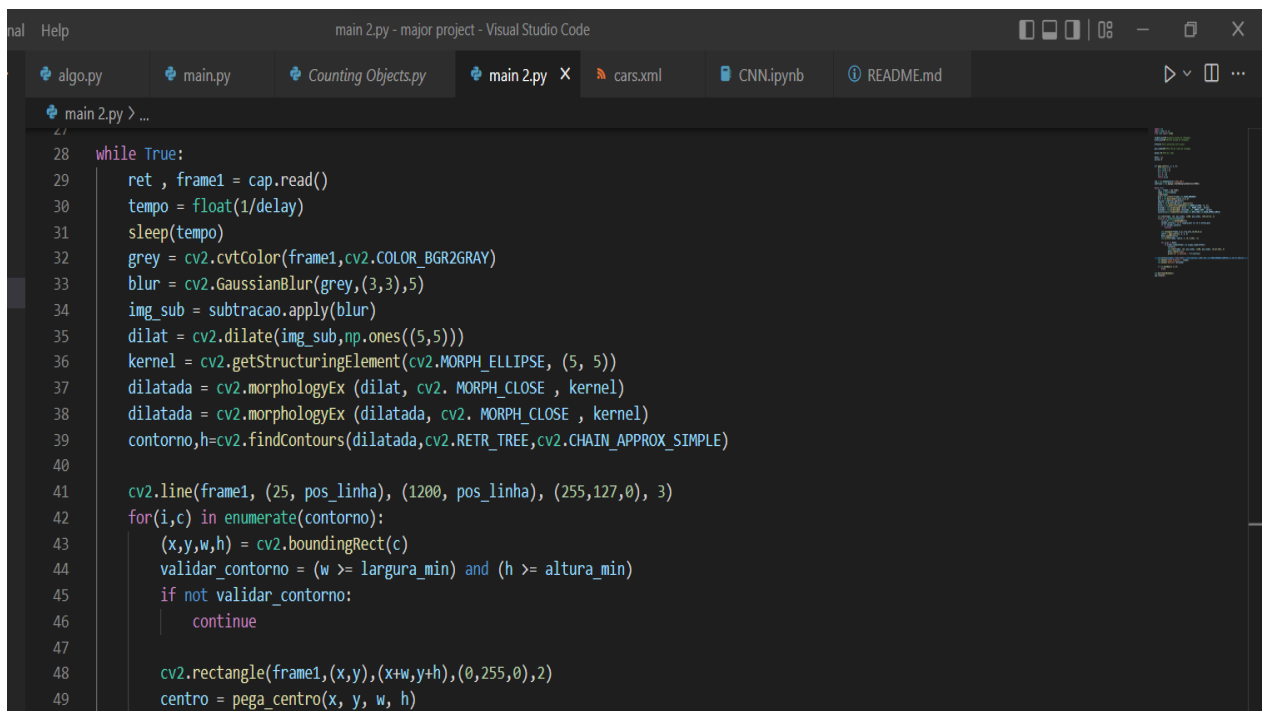


Figure 6.8 - Code

Chapter F. CONCLUSION AND FUTURE WORK

CONCLUSION

This study shows that object detection proposed an objective for tracking and counting when we amplify it on other real-time applications, so we can solve so many problems like traffic-related or any other where we need to be done and also, Haar-Cascade is very useful and it is only work for image detection, so machine learning for this project was chosen because of its advantages. And also, using these things, we can make so many other things happen which are futuristic, and also, using this, we can control accidents which happen because of poor traffic management, so in the end, this will be a thing which improves our future in real time incidents.

FUTURE WORK

Object detection, completely inter-linked with different comparable CV strategies, image segmentation, and image reputation that assist us to understand and analysis the scenes in motion pictures and pictures. The future of detection technology is within the process of proving itself and, similar to the original commercial revolution, it has the capability to free people from menial jobs that can be achieved more effectively and efficiently by using machines so that you can detect multiple objects. We want to create multiple instances of the correlation tracker item, after which while we need to compute the area of the object in subsequent frames, we need to loop over all n item trackers and take hold of the most recent role.

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